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ORIGINAL PAPER

Familial Educational Factors and COVID-19 Vaccine Beliefs Among Mexican-Origin Youth

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Abstract
Mexican-origin youth, a substantial and expanding demographic among U.S. youth, were disproportionately affected by COVID-19. Understanding the factors shaping their COVID-19 vaccine beliefs is crucial for informing future pandemic preparedness and vaccination promotion efforts targeting this vulnerable population. While previous research has established links between education and health literacy (i.e., COVID-19 vaccine beliefs), most studies focus solely on educational attainment. Given that most youth have not yet fully realized their educational attainment, this study further investigates how youth's educational expectations and parents' educational levels influence their COVID-19 vaccine beliefs. Using data from 229 Mexican-origin youth, along with 224 of their mothers and 100 of their fathers, this study assesses the impact of youth's educational expectations, educational attainment, and their parents' educational attainment on their COVID-19 vaccine beliefs including motivation, attitudes, information access, and hesitancy. Regression analyses suggests a significant influence of youth's educational expectations on their attitudes toward COVID-19 vaccines, surpassing the impact of the youths and their parents' educational attainment. These results underscore the importance of fostering and nurturing educational aspirations among minority youth. Future studies should acknowledge the multifaceted nature of education as a determinant of youth's COVID-19 vaccine beliefs, encompassing not only attainment but also educational expectations and parents' educational attainments.

Keywords Covid-19 vaccine · Mexican-origin youth · Educational attainment · Educational expectation · Parents' education

Background

The COVID-19 pandemic has revealed significant racial and ethnic disparities, with the Latino population being disproportionately affected. Compared to White populations, Latinos were three times more likely to become infected by COVID-19 and twice as likely to die from it [1]. Despite the critical role of COVID-19 vaccine in curbing the pandemic, their distribution has been inequitable, as areas with a larger Latino population were less likely to have access to COVID-19 vaccination facilities [2]. Such systemic barriers has contributed to negative beliefs about COVID-19 vaccines within Hispanic and Latinx communities, further hindering vaccine uptake [3]. This cycle of negative beliefs, decreased vaccine uptake, higher infection and death rates, and continued inequitable distribution can perpetuate a vicious circle of health disparities [4]. To address these systemic barriers and inform future emergency vaccination promotion efforts, it is crucial to examine factors influencing COVID-19 vaccine beliefs within Latino communities. Such effort is particularly important for Mexican-origin young adults, as the Mexican-origin population represents the largest subgroup within the Latino community [5]. Moreover, young adults are more likely to delay COVID-19 vaccination due to lower perceived severity of the disease and reduced perceived susceptibility to infection.

Education is an important determinant of health literacy including vaccination beliefs. According to the integrated

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model of health literacy [6], educational attainment influences how individuals access, understand, evaluate, and apply health-related information—such as information regarding COVID-19 vaccines. Youth with higher educational attainment tend to have greater knowledge and face fewer barriers to accessing COVID-19 vaccination information. They are also better equipped to understand scientific explanations regarding vaccine efficacy and safety. Consequently, these youth are more likely to be motivated to get vaccinated, and hold more positive attitudes and less hesitancy towards the vaccine. Previous studies found that highly educated youth reported greater acceptance of the COVID-19 vaccine [7] and were more willing to get vaccinated [8]. However, educational attainment alone may not fully explain the differences in how adolescents with the same level of education perceive or critically evaluate the COVID-19 vaccine. Previous studies suggest that other educational facets such as youth's educational expectations [9] and parents' education levels [10], may influence the way youth perceive the COVID-19 vaccine and how they access vaccine-related information. Stronger educational expectations and higher parental education are positively associated with youth's educational attainment. This aligns with broader research linking higher educational attainment to better health literacy and more favorable beliefs about COVID-19 vaccination. Furthermore, youth with higher educational expectations are often more future-oriented and goal-driven. Their focus on long-term success may enhance their motivation to adopt health-promoting behaviors such as vaccination, not only for their own safety but also for the well-being of the broader community. This future-oriented mindset can foster more positive beliefs about vaccination.

Thus, this study aims to extend the integrated model of health literacy within the context of COVID-19 by developing a deeper understanding of how various educational factors relate to youth's vaccine beliefs. Adopting a multidimensional approach, this study integrates previous findings on educational attainment, educational expectations, and parents' education to examine how these education facets are associated with COVID-19 vaccine beliefs.

Methods

Participants

The study draws on four waves of longitudinal data from Mexican-origin families in central Texas (Wave 1: 2012–2015, Wave 2: 2013–2016, Wave 3: 2017–2020, Wave 4: 2021–2023). Participants were recruited via school presentations, public records, and community outreach events. Surveys were presented in both English and Spanish and were administered by bilingual and bicultural interviewers. The final data include 229 Mexican-origin youths (M_{age}=20.54, SD=1.28, 61.6% are female, 74.2% are *U.S.* born), 224 of their mothers (61% finished middle school or have lower education level), and 100 of their fathers (62% finished middle high school or have lower education level). Please see Figure S1 in Supplemental for the data collection flowchart. Adolescents, mothers, and fathers provided informed consent for participation. Participating families were compensated $60 at Wave 1 and $70 at Wave 4. All procedures received Institutional Review Board approval from the University of Texas at Austin.

Measures

Demographic information included measures of youth gender, nativity, and Wave 4 age. Youth's educational expectations and educational attainment were also collected at Wave 4, ranging from 1 (Some middle school) to 8 (Finish a graduate degree (Master's degree, medical, etc.). Mothers' and fathers' educational attainments were collected in a previous Wave (2012–2015), ranging from 1 (No formal schooling) to 11 (Finished graduate degree (Master degree, medical, etc.)).

COVID-19 vaccine beliefs were measured at Wave 4, including vaccine motivation, positive attitude towards the vaccine, hesitancy toward the vaccine, and barriers to getting information about the vaccine. Vaccine motivation was measured by a single item "I am motivated to take the COVID-19 vaccine". Positive attitude toward the COVID-19 vaccine was measured by three items adapted from a previous study [11], ranging from 1 (strongly disagree) to 5 (strongly agree). Example items include "The COVID-19 vaccine can be trusted" and "The COVID-19 vaccine is safe." Higher mean scores represent more positive attitudes toward the COVID-19 vaccine (Cronbach's α=0.92). Barriers to getting information about the vaccine included three items, two of which were adapted from a previous study [12], and one item "I have a hard time finding information in Spanish on how to get the COVID-19 vaccine" (reverse coded) was developed for this study. All three items range from 1 (strongly disagree) to 5 (strongly agree). Higher mean scores indicate more knowledge of the COVID-19 vaccine (Cronbach's α=0.66). Hesitancy toward the vaccine included two items adapted from a previous study [13], ranging from 1 (strongly disagree) to 5 (strongly agree). Example item is included "I am hesitant (I have doubts) about taking the COVID-19 vaccine." Higher mean scores indicate higher level of COVID-19 vaccine hesitancy (Split-half reliability coefficient=0.83).

Analyses

All analyses were conducted using R. Correlation and regression analyses were conducted to explore the relationship from youth and parental education variables to COVID-19 vaccine-related beliefs. Four education variables were included in the regression model to account for potential interrelated effects among them. Youth gender, age, and nativity were also included in the regression model as covariates as previous studies indicated these factors may influence individuals' covid-19 vaccine beliefs [8].

Results

Table 1 displays participant demographics. According to the correlation analysis (see Fig. 1), educational variables positively correlated with COVID-19 vaccine motivation (r ranges from 0.20 to 0.34) and positive attitudes (r ranges from 0.25 to 0.36), while negatively correlated with barriers to accessing information (r ranges from −0.03 to 0.27) and vaccine hesitancy (r ranges from −0.13 to −0.22). Notably, relative to educational attainment variables, youth's educational expectations showed a more substantial correlation with COVID-19 vaccine variables (r ranges from −0.20 to

Table 1 Descriptive information of participants

	Overall (N=229)			
Youth gender				
Male	87 (38.0%)			
Female	141 (61.6%)			
Missing	1 (0.4%)			
Youth nativity				
U.S.	170 (74.2%)			
Mexico	59 (25.8%)			
Youth age				
Mean (SD)	20.5 (1.28)			
Median [Min, Max]	21.0 [17.0, 24.0]			
Motivation to vaccine One item '*I am motivated to take the COVID-19 vaccine'*				
Mean (SD)	3.00 (1.39)			
Median [Min, Max]	3.00 [1.00, 5.00]			
Positive attitude to vaccine Three items *i.e. 'The COVID-19 vaccine can be trusted'*				
Mean (SD)	3.42 (0.88)			
Median [Min, Max]	3.33 [1.00, 5.00]			
Barriers getting information about the vaccine Three items *i.e. 'Information about the COVID-19 vaccine is easy to access'*				
Mean (SD)	2.17 (0.66)			
Median [Min, Max]	2.00 [1.00, 4.00]			
Hesitancy of Covid-19 vaccine Two items *i.e. 'I am hesitant (I have doubts) about taking the COVID-19 vaccine'*				
Mean (SD)	2.46 (1.18)			
Median [Min, Max]	2.50 [1.00, 5.00]			
Education	***youth educational expectation***	***youth educational attainment***	***mother educational attainment***	***father educational attainment***
No formal schooling	0	0	9 (3.9%)	8 (3.5%)
Some elementary school	0	0	25 (10.9%)	16 (7.0%)
Finished elementary school	0	0	33 (14.4%)	12 (5.2%)
Some middle school/junior high school	0	0	28 (12.2%)	9 (3.9%)
Finished middle school/junior high school	0	1 (0.4%)	41 (17.9%)	17 (7.4%)
Some high school	3 (1.3%)	10 (4.4%)	29 (12.7%)	8 (3.5%)
Finished high school	34 (14.8%)	163 (71.2%)	24 (10.5%)	12 (5.2%)
Finished technical/vocational training after high school	27 (11.8%)	25 (10.9%)	19 (8.3%)	10 (4.4%)
Finished community college degree (AA)	33 (14.4%)	20 (8.7%)	4 (1.7%)	2 (0.9%)
Finished university/bachelor degree (BA/BS)	77 (33.6%)	5 (2.2%)	12 (5.2%)	5 (2.2%)
Finished graduate degree (Master degree, medical, etc.)	51 (22.3%)	0	0	1 (0.4%)
Missing	4 (1.7%)	5 (2.2%)	5 (2.2%)	129 (56.3%)

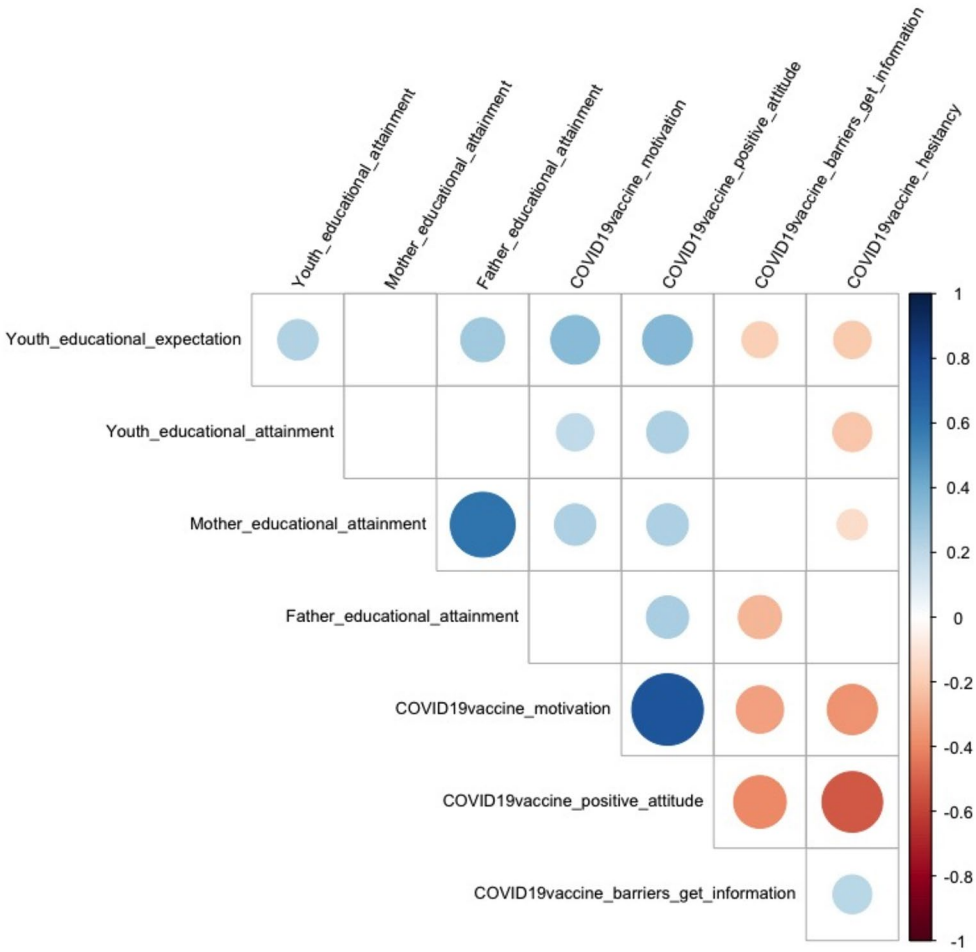

Fig. 1 Correlations among four education variables and COVID-19 vaccine beliefs. Colored dots in cells indicate significant findings. Blue color indicates a positive correlation; red color indicates a negative correlation

0.36). Mothers' and fathers' educational attainment also correlated to different aspects of COVID-19 vaccine beliefs. More specifically, mothers' educational attainment significantly correlated with motivation to COVID-19 vaccine ($r=0.24$, $p<0.001$), positive attitude to COVID-19 vaccine ($r=0.25$, $p<0.001$), and hesitancy of COVID-19 vaccine ($r=-0.13$, $p<0.05$). Fathers' educational attainment significantly correlated with positive attitude to COVID-19 vaccine ($r=0.26$, $p<0.01$), and barriers to getting information about COVID-19 vaccine ($r=-0.27$, $p<0.01$).

Regression results (see Table 2) indicated that when considering all education variables together, only youth's educational expectation had a significant influence on youth's COVID-19 vaccine motivation ($b=0.326$, $SE=0.114$, $p=0.005$), positive attitude ($b=0.267$, $SE=0.067$, $p<0.001$), hesitation ($b=-0.204$, $SE=0.093$, $p=0.031$), and barriers of getting information ($b=-0.169$, $SE=0.055$, $p=0.003$). The variance inflation factor (VIF) scores were in a range of 1.03 to 1.63 (see Supplemental Table S1 for VIF scores of variables included in regression models), indicating an acceptable level of multicollinearity among the educational variables [14].

Discussion

The findings add to research on COVID-19 vaccination by contributing a multidimensional perspective in understanding the role of education in shaping Mexican-origin youth's COVID-19 vaccine beliefs, who are uniquely vulnerable to COVID-19.

Educational Expectations and Vaccine Beliefs

The results suggest that educational expectations may be the most critical and robust educational factor informing Mexican-origin youth's vaccination beliefs; youth with higher educational expectations may possess more optimistic expectations for their future and thus view COVID-19 vaccination as a means to not only safeguard their concurrent health, but also their prospects for a successful future. This intrinsic motivation could foster favorable attitudes towards the COVID-19 vaccine. These results underscore the importance of fostering and nurturing youth's educational aspirations, as they can have far-reaching implications for youth's beliefs regarding immunization initiatives such as

Table 2 Regression of educational factors on Covid-19 vaccine beliefs

	Motivation to Covid-19 vaccine			Positive attitude to Covid-19 vaccine			Barriers getting information about Covid-19 vaccine			Hesitancy of Covid-19 vaccine		
	b	95% CI	*p*	b	95% CI	*p*	b	95% CI	*p*	b	95% CI	*p*
Age	0.04	[-0.21, 0.28]	0.75	0.07	[-0.07, 0.21]	0.34	-0.04	[-0.16, 0.08]	0.50	-0.10	[-0.30, 0.10]	0.30
Gender (0 = male, 1 = female)	0.47	[-0.12, 1.06]	0.12	0.04	[-0.31, 0.38]	0.83	0.1	[-0.19, 0.38]	0.51	-0.05	[-0.53, 0.43]	0.84
Nativity (0 = US, 1 = Mexico)	-0.21	[-0.88, 0.46]	0.53	-0.20	[-0.60, 0.19]	0.31	-0.09	[-0.41, 0.24]	0.59	0.17	[-0.38, 0.72]	0.54
Youth educational expectation	**0.33**	**[0.10, 0.55]**	**<0.01**	**0.27**	**[0.13, 0.40]**	**<0.01**	**-0.17**	**[-0.28, -0.06]**	**<0.01**	**-0.20**	**[-0.39, -0.02]**	**0.03**
Youth educational attainment	0.29	[-0.18, 0.76]	0.22	0.15	[-0.13,0.43]	0.28	-0.15	[-0.38, 0.08]	0.19	-0.15	[-0.53, 0.24]	0.45
Mother educational attainment	0.09	[-0.06, 0.23]	0.24	0.06	[-0.03, 0.14]	0.18	0.04	[-0.03, 0.11]	0.22	-0.01	[-0.13, 0.11]	0.86
Father educational attainment	0.02	[-0.12, 0.16]	0.77	0.03	[-0.05, 0.11]	0.44	-0.06	[-0.13, 0.00]	0.06	-0.05	[-0.16, 0.07]	0.41
Adjusted R-square	0.14			0.23			0.15			0.04		

Note. Significant findings are in bold

the COVID-19 vaccine. The implications of these findings extend to policymakers and educational institutions. There is a clear need to enhance educational outreach for minoritized youth by providing information on pathways to higher education, academic opportunities, and financial aid resources. This is particularly important because these youth often face contextual stressors, such as discrimination, which can limit access to higher education opportunities and lower their educational expectation [15]. These negative experiences can also contribute to negative vaccine beliefs and mistrust in healthcare institutions. To address these challenges, targeted college-level outreach programs should be implemented in schools with large populations of low-income and ethnic-minority students. From a clinical perspective, these results offer valuable insights for vaccine promotion. Clinicians and health professionals can tailor communication strategies for minoritized youth by connecting the benefits of vaccination to their educational and career goals. Aligning health messages with youths' expectation can encourage vaccination, promoting both individual well-being and community health.

The current study primarily focuses on investigating how different facets of educational factors—youth educational attainment, educational expectations, and parental educational attainment—concurrently affect COVID-19 vaccine beliefs. However, youth's educational expectations may change over time, potentially reflecting their academic performance, shifts in family socioeconomic status, and evolving perspectives about the future. These changes in educational expectations could influence their future beliefs about the COVID-19 vaccine. Therefore, future studies should consider examining the trajectory of youth educational expectations from early adolescence to emerging adulthood and explore how these changes might influence their vaccine beliefs.

Parental Education and Youth's Vaccine Beliefs

Correlational results reveal the possible association between mothers' and fathers' educational attainment on youth's COVID-19 vaccine beliefs. Parents' educational attainment may shape youth perceptions of the vaccine and influence how parents communicate vaccine-related information to their children, indirectly affecting youth's vaccine beliefs. This effect may be more pronounced in Mexican-origin families compared to White families, as traditional Mexican culture emphasizes family, parental authority, and collectivism, making parents' beliefs likely to be passed down to their children [16]. However, this study did not directly examine parents' COVID-19 vaccine beliefs or parent-child communication, highlighting an important area for future research. Moreover, parents who attain higher education can

serve as role models, inspiring youths to develop intrinsic motivation to pursue further education, and consequently influence youth's COVID-19 vaccine beliefs. Therefore, future research should explore whether adolescents' educational expectations mediate the relationship between parents' educational attainment and youths' beliefs about COVID-19 vaccines to further explain the potential mechanisms by which parental educational attainment influences COVID-19 vaccine beliefs.

Limitations

The current study has several limitations. First, the relatively small sample size of fathers necessitates caution in interpreting the results. Second, the majority of participating parents had completed middle school education or lower, and most families were from low-income backgrounds. Given the heterogeneity in socioeconomic status among Mexican-origin families, the findings from this study may not be generalizable to all Mexican-origin families across the United States. Additionally, as this is a descriptive and exploratory study, the interactive effects of different educational variables were not tested. As youth and their parents from low-income Mexican-origin families may have different levels of educational attainment, resulting in varying vaccine beliefs, future research should explore the potential interactive effects of both youth and parental educational attainment to better understand the complexities of family education dynamics on youth vaccine beliefs. Investigating these interactions is crucial for developing tailored intervention strategies for vaccine promotion efforts, particularly for youth with diverse personal and familial educational backgrounds.

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Declarations

Ethical Approval This study was performed in line with the principles of the Declaration of Helsinki. Approval was granted by the Institutional Review Board at the University of Texas at Austin, protocol number 2015-01-0006.

Consent To Participate Informed consent was obtained from all individual participants included in the study.

Competing Interests The authors report there are no competing interests to declare.

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